Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011

Electronics I

Experiment (2) Introduction to LTspice 2

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Experiment (2)
Introduction to LT Spice

Objectives

- To simulate a first-order low-pass filter circuit using AC analysis.
- To be able to present results using plots for various circuit variables.

Pre-lab Work

1) Before starting this experiment, you need to study the document "Introduction to LT Spice IV with Examples", by Nicholas Lombardo, Linear Technology, 2015.
2) Analyze the circuit in figure 1. In particular, find the frequency response of the transfer function and filter bandwidth.

Circuits

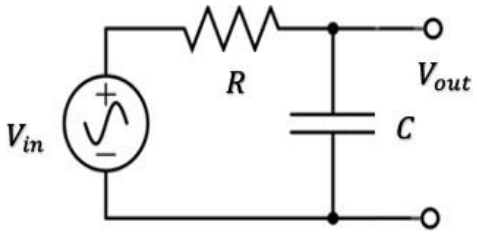


Figure 1: RC low-pass filter.

Procedure

Part (1): AC Analysis.

1. Connect the circuit in figure 1. Use the circuit elements values given by the lab instructor. V_{in} is a sinusoidal AC voltage source with frequency f and maximum amplitude V_{max}.
2. Simulate the circuit using the AC analysis function and generate the transfer function spectrum curve (versus frequency). Use dB scale for the transfer function magnitude and log scale for the frequency.
3. Change the value of C and observe the simulation curves.

Results and Analysis

Part (1): AC Analysis.

1. From your simulations, find the simulated bandwidth of the filter for different values of C. Record your results in table 1.

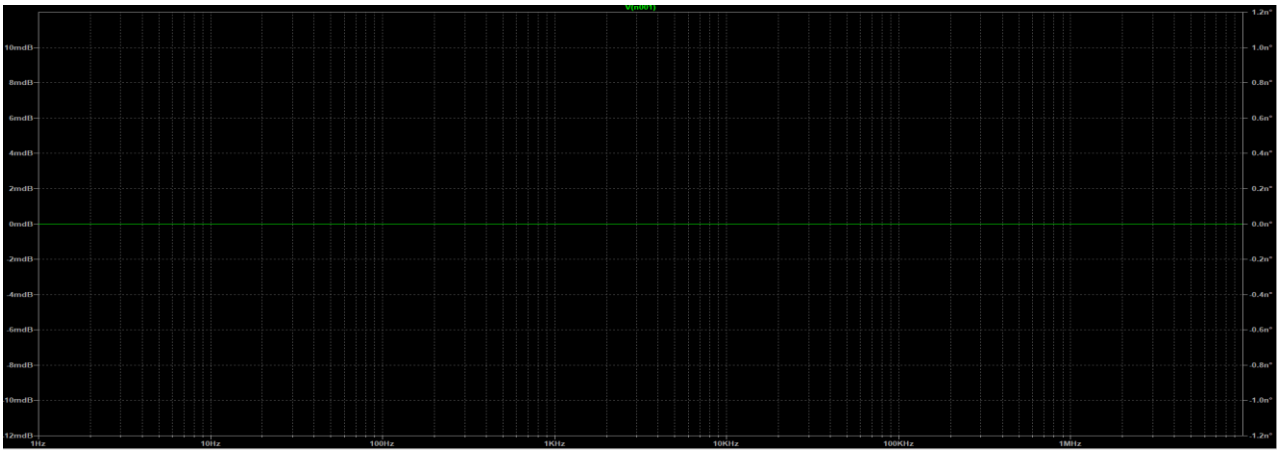
Table 1. Simulated and calculated bandwidth.

Capacitor value	Simulated Bandwidth	Theoretical Bandwidth	% Difference
100nF	1.59327 kHz	1.58 kHz	0.83%
10nF	15.9327 kHz	15.8 kHz	0.83%
1nF	159.327 kHz	158 kHz	0.83%

2. Comment on the results in table 1.

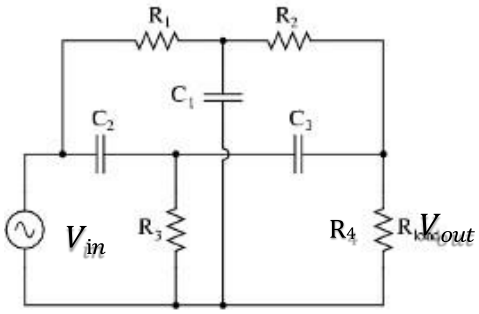
Since the error was very small, the same results were obtained for 3 different capacitor values. Since the difference between simulation and calculated values is very small, this difference can be neglected.

3. If the output of the circuit is taken across the resistor instead of the capacitor, what kind of a filter is obtained? Explain your answer and support it using simulation results.

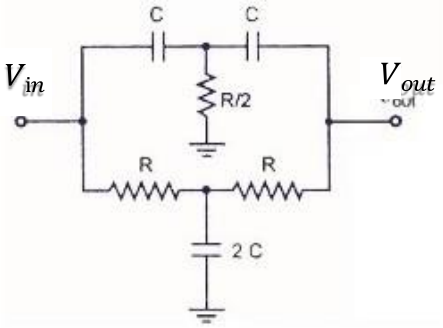


There is no change in frequency

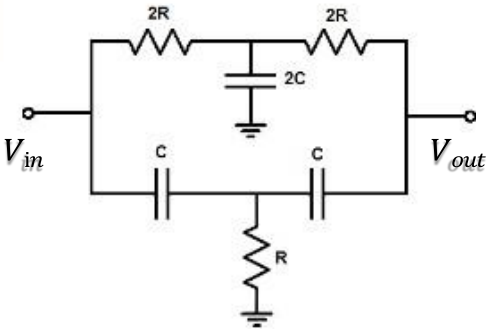
Lab Assignment



Circuit 1



Circuit 2



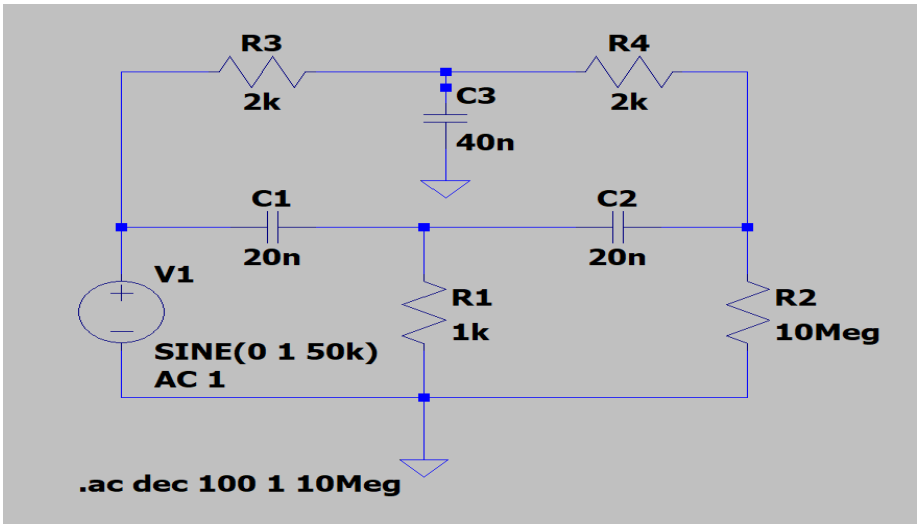
Circuit 3

1. The lab instructor will assign one circuit to you with appropriate elements values.
2. Connect the circuit using the circuit elements values given by the lab instructor. V_{in} is a sinusoidal AC voltage source with frequency f and maximum amplitude V_{max}.
3. Simulate the circuit using the AC analysis function and generate the transfer function spectrum curve (versus frequency). Use dB scale for the transfer function and log scale for the frequency.
4. From your simulations, find the simulated bandwidth of this filter circuit.
5. Change the value of the circuit elements (one at a time) and observe the simulation curves. Write your conclusions regarding the effect of these elements on the circuit bandwidth. Can you come-up with a formula for the bandwidth in terms of C and R?
6. What kind of a filter is this circuit? Explain.

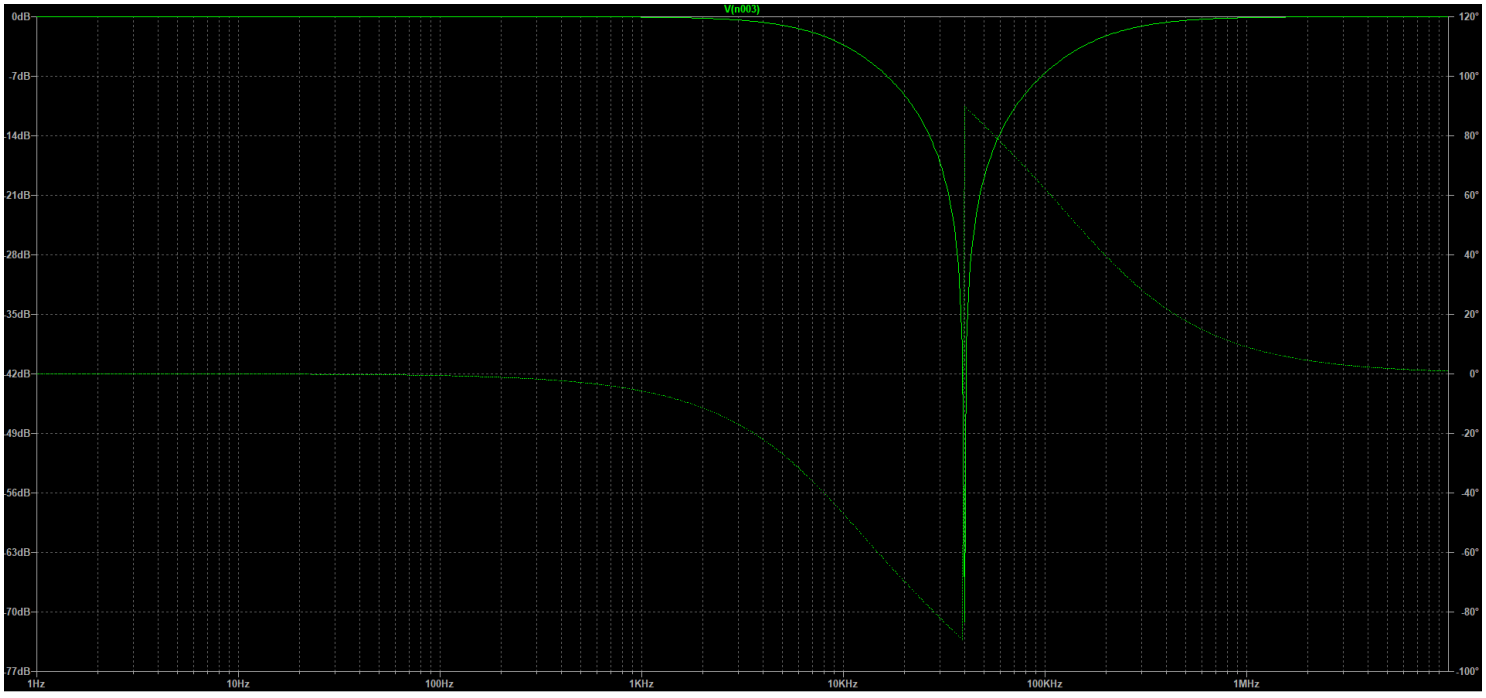
Part II

1) Our lab assistant gave us the first circuit diagram as a task.

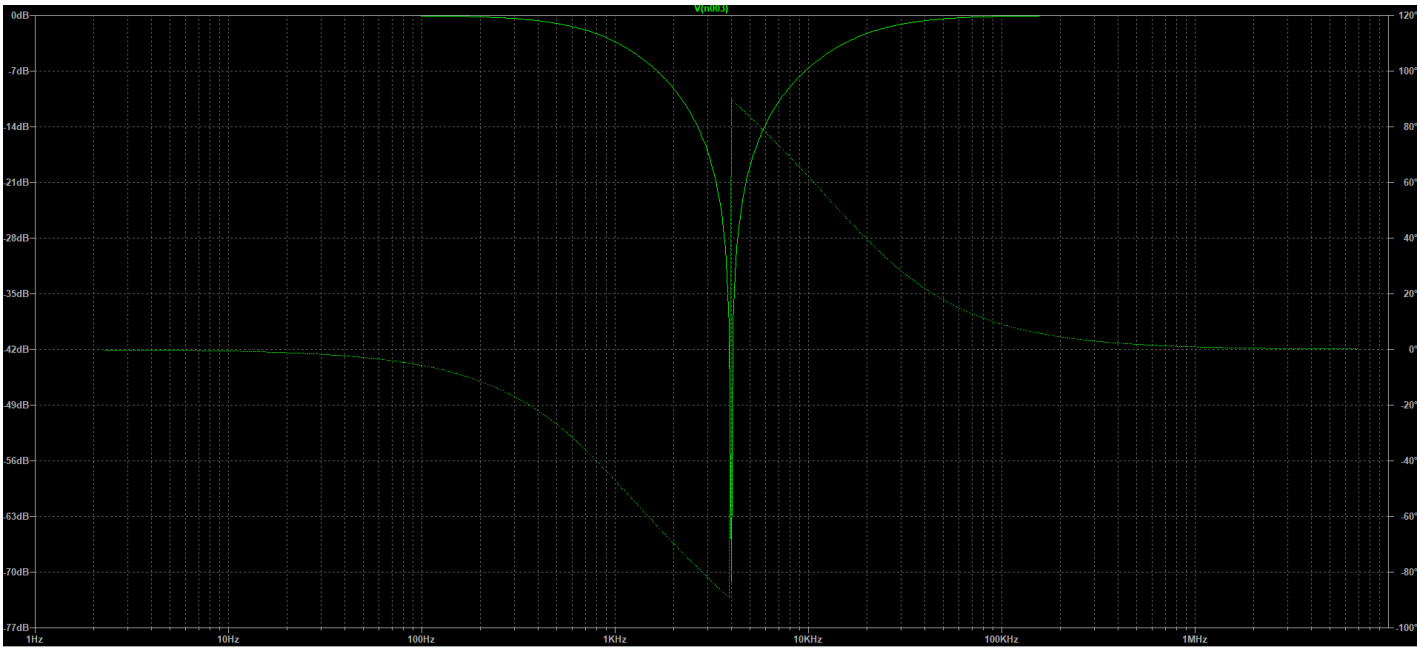
2)



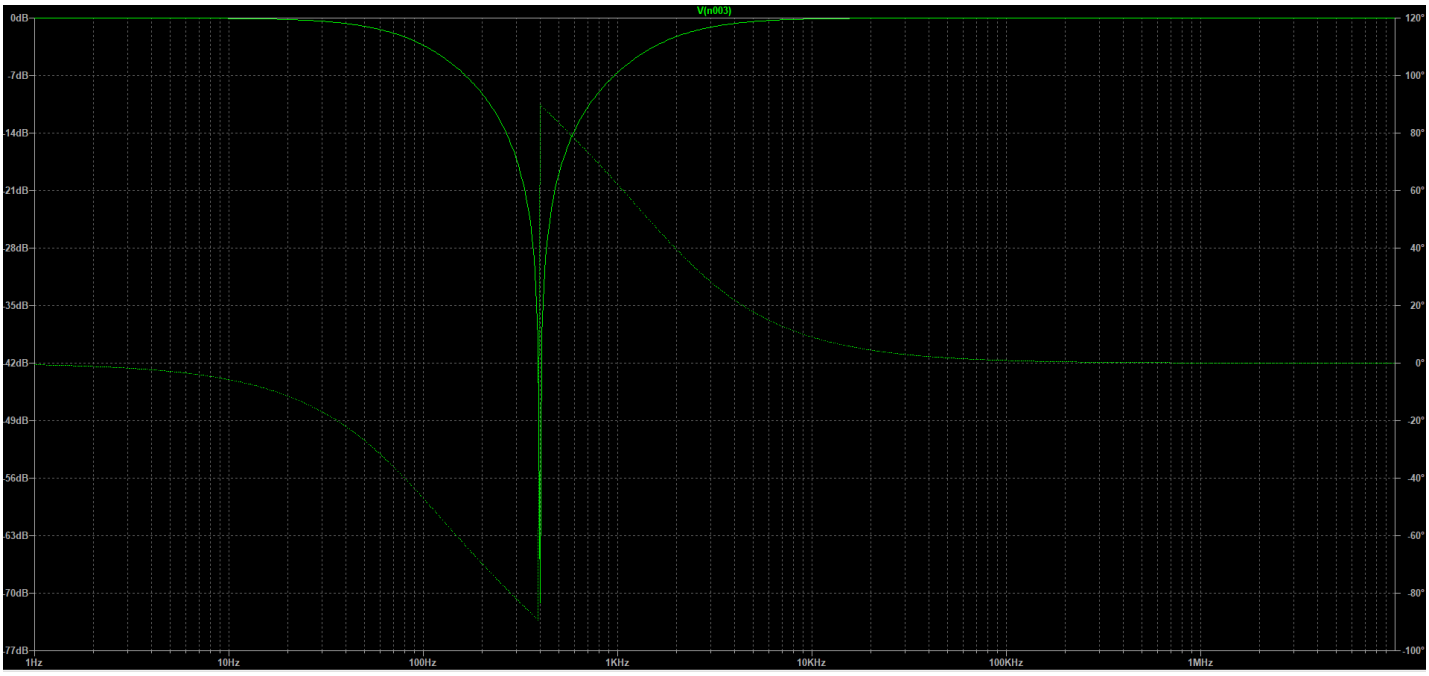
2) $c_1 = 2n$, $c_2 = 2n$, $c_3 = 4n$



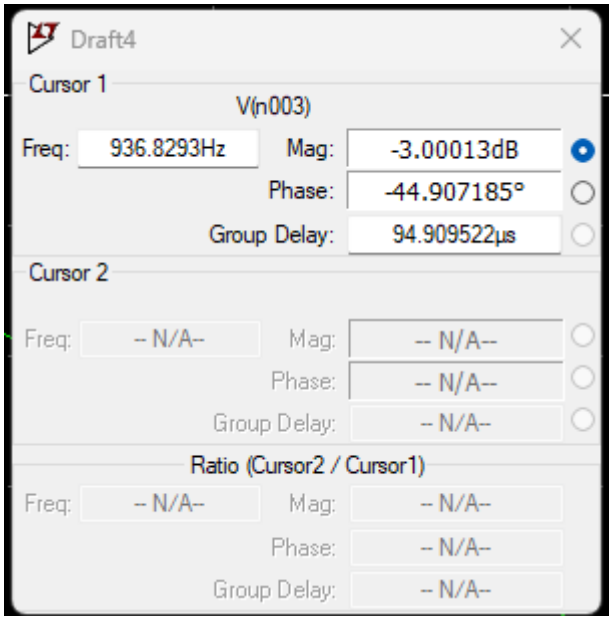
$c_1 = 20n, c_2 = 20n, c_3 = 40n$

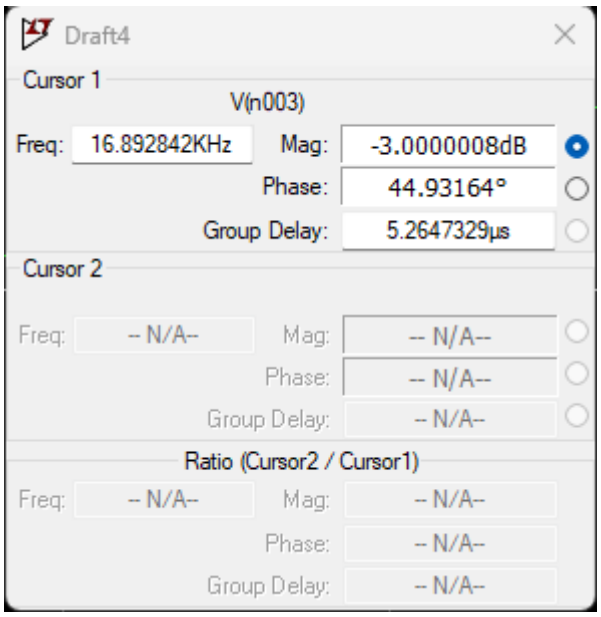


$c_1 = 200n, c_2 = 200n, c_3 = 400n$



4)





The bandwidth for this circuit is (16892.8)Hz – (936.8)Hz = 15.9 KHz.

5) If we change the value of only 1 element at a time, the bandwidth changes, but if we change only the capacitor or resistor values at a time, the bandwidth does not change.

6) This filter is a Band-stop filter. While it normally passes all other frequencies except the frequency in a certain range, it greatly reduces the magnitude of the frequency in the range.